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***Development of a Mobile Application for Daily
Diabetes Management***

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General introduction

Diabetes is not just a medical diagnosis, it's a lifelong journey. From the moment of diagnosis, daily life changes in ways that are often invisible to others. Managing diabetes requires constant blood sugar monitoring, careful meal planning, regular physical activity, and adherence to medical appointments, all while striving to maintain a sense of normalcy.

This project was initiated with a clear mission: to leverage the skills acquired during our computer science studies to develop an application that alleviates some challenges faced by individuals living with diabetes. The primary objective was to build a mobile app offering smarter, personalized support for people with diabetes. Additionally, we aimed to foster a strong connection between patients and doctors. We hope this will improve efficiency and accessibility in diabetes care.

Our application was designed to accommodate various patient profiles, including those managing type 1 diabetes, type 2 diabetes, gestational diabetes and prediabetes. Key features include real-time alerts for abnormal glucose levels, medication and appointment reminders, and dedicated interfaces for doctors that will help their patient and be more present even though the distance between them.

To ensure the app works across devices and is accessible to as many users as possible, we developed it using Dart and Flutter, a cross-platform framework that allows for smooth performance on both Android and iOS.

This thesis documents the entire development process, from our initial research and design to implementation, testing and evaluation of the app's potential real-world impact. The iterative nature of the project allowed us to refine the application and improve its usability.

Ultimately, our hope is that this app can go beyond just tracking data. We want it to feel like a companion that helps users take control of their condition, feel less alone, and stay one step ahead in their health journey. However, as diabetes management remains complex and often overwhelming, particularly without constant medical support, many patients struggle to track their condition effectively, risking serious complications such as heart diseases, neuropathy, and other diseases. How can mobile technology be leveraged to provide personalized, accessible, and supportive care for diabetic individuals in their day-to-day lives?

This thesis is structured into three main chapters. The first chapter sets the foundation by presenting the project context, defining the problem, and detailing the research methodology, including data collection and expert consultations. The second chapter focuses on analyzing the needs and modeling the system through detailed requirements and diagrams that guide the design process. Finally, the third chapter covers the implementation phase, describing the development tools, technologies, and frameworks used, as well as the application's interfaces.

Chapter 1: Project context and preliminary analysis

1.1 Introduction

This chapter provides a comprehensive overview of the project's background and essential preliminary analyses. It introduces the research context, highlights the main challenges faced in diabetes management, presents the objectives of the current study, and outlines the methodology used for gathering user insights. The chapter also defines the target audience and includes an analysis of existing applications and user expectations.

1.2 Research Context and Problem Statement

Diabetes is a chronic disease that affects millions of people worldwide. The rising prevalence of diabetes presents significant challenges for both patients and healthcare providers. Effective diabetes management requires continuous monitoring of blood sugar levels, adherence to a proper diet, regular exercise, and timely medical consultations. However, many patients find it difficult to track their health data and make informed decisions regarding lifestyle and treatment.

In recent years, technological advancements have played a significant role in improving healthcare solutions. Mobile applications, in particular, have emerged as valuable tools for managing chronic diseases like diabetes. These applications offer features such as blood sugar tracking, medication reminders, and doctor consultations, making it easier for patients to stay on top of their health and effectively manage their condition on a daily basis.

1.3 Study Overview, challenges, and objectives

This thesis presents the development of a mobile application designed to assist individuals with diabetes in effectively managing their health. The application aims not only to provide the necessary functionalities for daily management, but also to facilitate communication with healthcare providers. Ultimately, the goal is to help users achieve and maintain stable blood sugar levels, thereby preventing acute and chronic complications related to diabetes. To ensure that the application meets the needs of its target users, a survey was conducted to collect accurate data on the challenges faced by people with diabetes and their expectations from a management tool. The insights gained from this survey have been instrumental in shaping the features and functionality of the application.

1.4 Target Audience

Our application is designed for two main user groups:

Patients with diabetes and healthcare professionals. The goal is to assist individuals in managing their condition effectively while also providing doctors with relevant patient data to facilitate better treatment decisions. The patient group includes individuals with different forms of diabetes, each requiring specific management strategies:

Type 1 Diabetes: A chronic autoimmune disease in which the immune system destroys the insulin producing beta cells in the pancreas, resulting in little to no insulin production. Individuals with this condition must manage their blood glucose through daily insulin injections or an insulin pump.[1]

Type 2 Diabetes: A long-term metabolic condition where the body either doesn't respond properly to insulin (insulin resistance) or doesn't produce enough of it. Management typically involves a combination of healthy lifestyle habits, oral medications, and sometimes insulin.[1]

Gestational Diabetes: A temporary form of diabetes that develops during pregnancy, characterized by elevated blood sugar levels. It generally resolves after childbirth but increases the risk of developing type 2 diabetes later in life. Management includes a healthy diet, physical activity, and, in some cases, medication.[1]

Prediabetes: A condition where blood glucose levels are elevated above normal but not high enough for a diabetes diagnosis. Without intervention, it often progresses to type 2 diabetes, but it can be reversed through lifestyle changes like improved diet and increased physical activity.[1]

By addressing the needs of these diverse users, the application aims to provide tailored support for diabetes management, improving both self-care for patients and decision making for doctors.

1.5 Methodology and Data Collection

1.5.1 Survey and User Needs Assessment

To ensure the application aligns with the real needs of its users, we conducted a comprehensive survey targeting individuals with diabetes. The goal was to gather detailed insights into their daily management routines, treatment methods, challenges, and attitudes toward mobile health applications. By distributing the form to a diverse group of patients, we aimed to assess both their current experiences and their willingness to use a digital platform for sharing health data. The results of this survey played a crucial role in shaping the app's features and overall functionality.

A total of 654 responses were collected by distributing and collecting data through an online form over a span of 24 hours. The majority of participants were between the ages of 46 and 60 (**31%**) and were primarily diagnosed with type 1 diabetes (**45%**), though a significant proportion had type 2 diabetes (**40%**). Regarding treatment methods, **52%** relied on oral medication, while **48%** used insulin. In terms of medical follow-up, **45%** reported consulting their doctor every three to six months.

Diabetes significantly impacted participants' health; **52%** reported experiencing severe hypoglycemic or hyperglycemic episodes that required hospitalisation. Lifestyle habits also presented concerns, as **38%** consumed high-sugar foods daily, and **43%** reported rarely engaging in physical activity. Importantly, **97%** of respondents did not use any mobile applications to manage their diabetes. However, **70%** expressed a willingness to share their health data through a dedicated digital tool, highlighting a significant gap in the adoption of existing applications.

The Following link provides access to the form: [diabetes form](#)

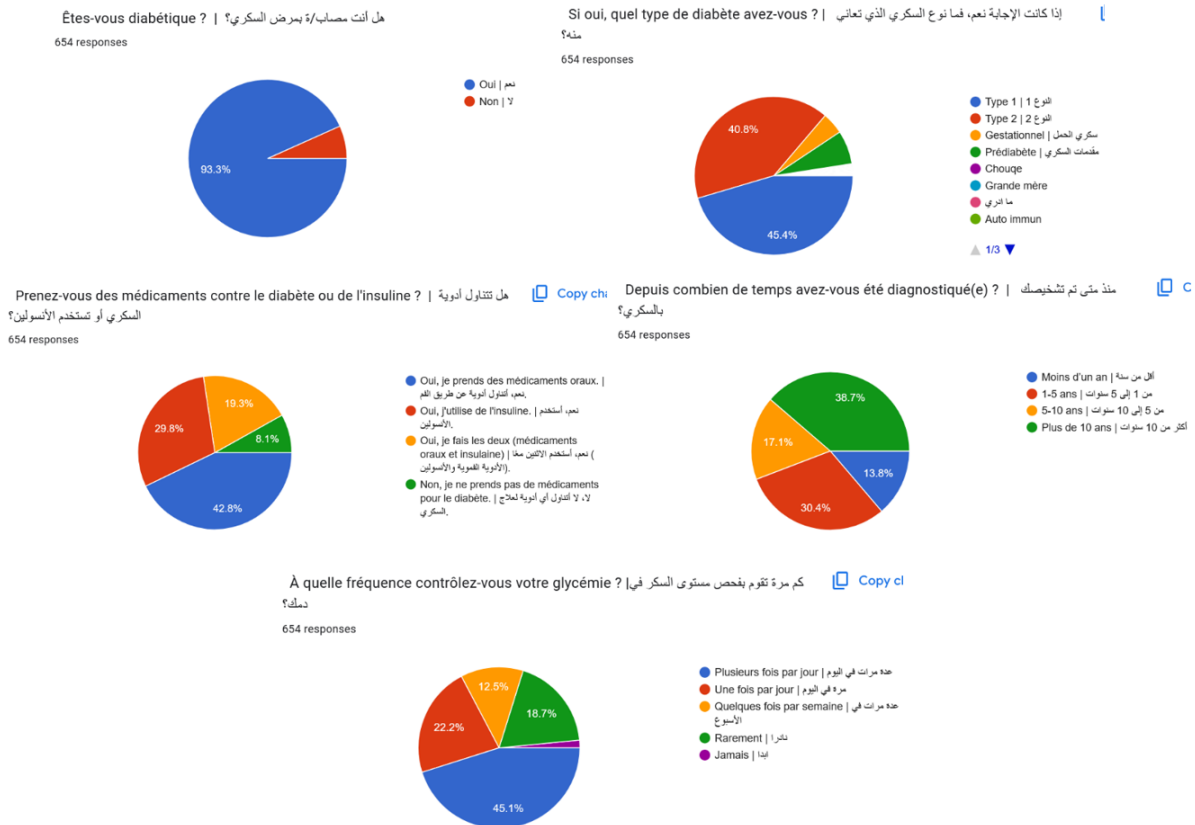


Figure 1: Screenshot of the form

1.5.2 Analysis of Existing Applications and User Expectations

Only 2.4% of respondents reported using a mobile app to track their diabetes. Among the applications mentioned were *Freestyle*, *LibreLink*, *Sibionics*, *MyTherapy*, *Contour*, *GluciCheck*, and *Medtronic Pump*. Those who did not use such tools cited a lack of awareness or the absence of applications that met their specific needs.

Despite the variety of existing applications mentioned by respondents, our analysis revealed several common shortcomings. Most of these tools focus on a narrow set of features, primarily centered around blood glucose monitoring, without offering a comprehensive approach to diabetes management. Few provide personalized nutritional guidance, tailored support for different types of diabetes, or user-friendly interfaces for healthcare professionals. Additionally, the lack of community spaces or psychological support features limits user engagement. These gaps highlight the need for a more complete, personalized, and interactive solution which is precisely what our application aims to deliver.

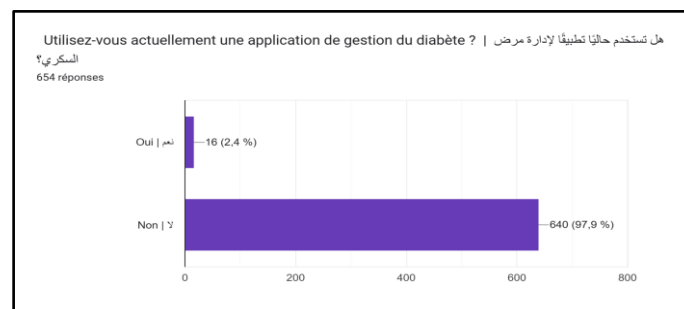


Figure 2 : Screenshot of percentage of diabetes app users

To gain further insight, we analysed the features of these applications, evaluating their benefits and limitations while identifying areas for improvement. Additionally, we asked participants about the functionalities they would find most valuable in a diabetes management app. The results indicated strong interest in the following features:

- Blood glucose monitoring and alerts (72%)
- Meal planning assistance (56%)
- Medication reminders (47%)
- Medical and emergency contact integration (29%)
- Community support and chat features (18%)

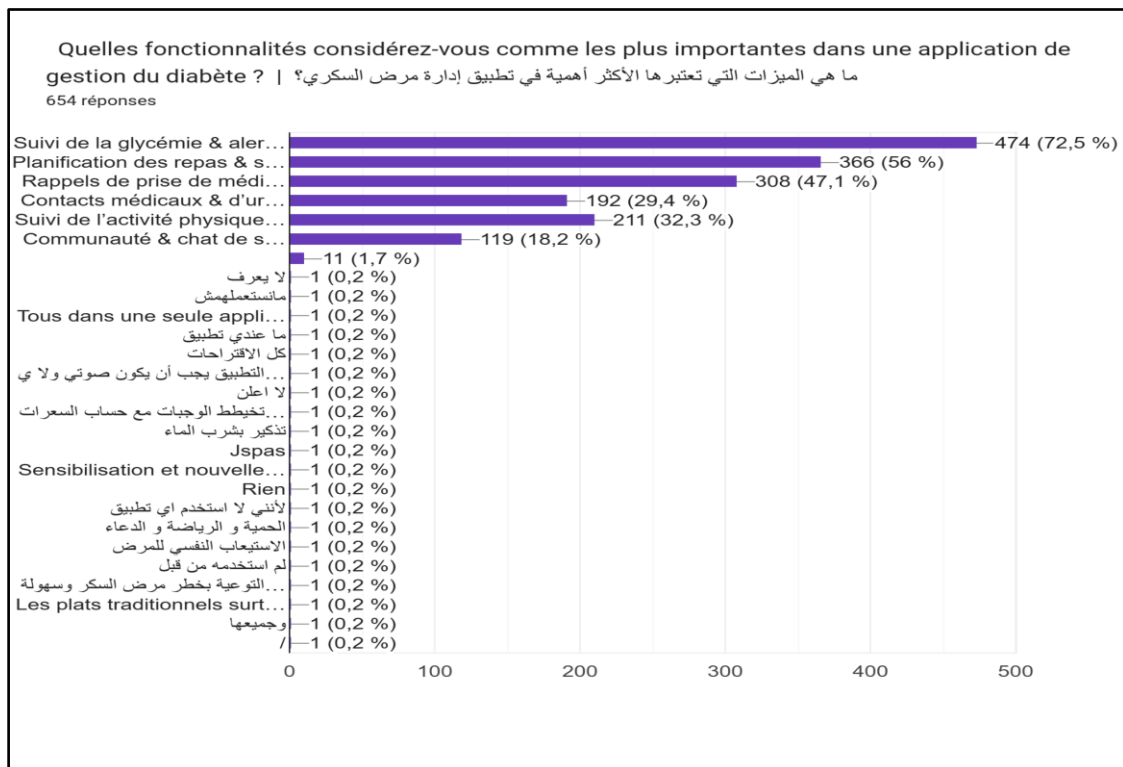


Figure 3 : Screenshot of patient's most valued app functionalities.

1.6 Medical Consultation and Expert Opinions

To ensure that our application aligns with medical best practices, we consulted with healthcare professionals. These discussions provided valuable insights into the challenges faced by patients and the key features that would make a diabetes management app more effective from a doctor's point of view. Doctors emphasized the importance of real time blood glucose monitoring with automated alerts to help prevent dangerous fluctuations. Another highly requested feature was a medication tracking system with reminders to improve adherence to prescribed treatments.

Furthermore, medical professionals highlighted the psychological impact of diabetes, particularly among younger patients who struggle to accept their diagnosis. They suggested incorporating a peer support network where patients can exchange experiences, seek advice, and stay motivated in managing their condition.

Finally, doctors stressed the need for an integrated medical follow up system that allows patients to log and share their health data with their physicians. This would facilitate more informed consultations and enable healthcare providers to detect potential complications at an early stage.

1.7 Conclusion

The findings of this survey, combined with expert input from medical professionals, reinforce the need for a more comprehensive and user-friendly diabetes management application. The willingness of 70% of respondents to share their health data suggests an opportunity to enhance personalization and improve patient outcomes. These insights will guide the development of our application, with a strong focus on ease of use, tailored recommendations, and community support.

Having presented the context, challenges, objectives, user needs, and the limitations of existing solutions, the next chapter will focus on analysing user needs and defining the system's functional structure through requirements, use case diagrams, and class modelling.

Chapter 2: Needs analysis and general modelling

2.1. Introduction

Mobile technology has revolutionised healthcare, providing effective tools for managing chronic conditions like diabetes. These tools facilitate continuous monitoring of blood sugar, dietary habits, and medication adherence. To address this need, we propose a diabetes management application designed to empower users to maintain their health through real-time monitoring, personalised guidance, and seamless communication with healthcare providers.

This chapter outlines the core requirements of the system, providing a clear framework for its development. Our approach emphasises a thorough understanding of both the system's functionalities and its performance criteria.

By defining functional and non-functional requirements, we establish the foundation for a user-centred and effective application. We employ established methodologies and technologies, such as the Unified Modelling Language, using use case, class, and sequence diagrams to model the system's functionality, structure, and relationships. This approach ensures a clean, intuitive, and scalable design.

2.2 Functional and Non-Functional Requirements

To ensure the application effectively meets user needs, it is crucial to define both functional and non-functional requirements that facilitate seamless and effective diabetes management.

2.2.1 Functional Requirements

Functional requirements define the specific tasks the application must perform to meet the core needs of its users. Key functionalities for this system include:

- **User authentication and account management:** The application will allow users to securely create and manage accounts
- **Blood sugar monitoring:** Users can manually enter their blood sugar levels throughout the day, enabling continuous tracking. Healthcare providers can then review these records and prescribe the insulin dosage the patient needs to adhere to and provide informed medical advice.
- **Patient/doctor interactions:** The application will facilitate communication between patients and healthcare providers through secure chat, health data sharing and optional virtual consultations for more comprehensive care.
- **Health reports and analytics:** The system will generate detailed health reports including glucose trends, dietary habits, and physical activity. Users can view interactive charts and graphs to monitor their progress. In addition, doctors can access these reports to inform their medical guidance and treatment recommendations.
- **Medication reminders:** Users can set up automated reminders for insulin and other medications, with the system delivering notifications to ensure adherence to their prescribed regimen via push notifications and SMS messages.
- **Emergency alerts:** If a patient's blood sugar levels reach dangerously high or low levels, the system will trigger an emergency alert, notifying a designated emergency contact or healthcare provider and sharing the patient's real-time location to facilitate a faster response.

2.2.2 Non-Functional Requirements

In addition to the core functionalities, the success of this diabetes management application depends on meeting key quality attributes[4]. These non-functional requirements ensure a reliable, secure, and user-friendly experience, and include but are not limited to:

- **Security and privacy:** The security and privacy attribute focus on the ability to safeguard the application, patients' data and information from unauthorized entities. This is very crucial as the data leaks may incur huge losses for both the patients and healthcare providers. Planned measures include data encryption, strong user authentication, and strict access control.
- **Reliability:** Reliability ensures the application performs as expected, even under peak loads, with minimal downtime, thus supporting continuous monitoring and care
- **Performance:** Performance refers to the application's ability to process and display health data in real time with minimal latency, relative to the resources available.
- **Compatibility:** Compatibility ensures the application functions seamlessly across various operating systems (iOS, Android) and browsers without compromising functionality.
- **Scalability:** Scalability refers to the system's ability to handle increased user load and data volume without impacting performance.
- **Usability:** The usability attribute refers to the end user's ease of use. Our app would have a user-friendly interface that is intuitive and fairly easy to navigate for users of all age groups.

2.3 The methodologies and technologies employed

2.3.1 Presentation of the technologies used

To design and model such an application, we rely on what could be considered standardised methodologies and tools. One of them is UML (Unified Modelling Language) used to visualise the system's architecture and user interactions.

Definition of Unified Modeling Language:



The Unified Modelling Language (UML) is a general-purpose visual modeling language that is intended to provide a standard way to visualize the design of a system, providing a standard notation for many types of diagrams, which can be roughly divided into three main groups: behavior diagrams, interaction diagrams, and structure diagrams [5]. We chose to use the use case diagrams that fall under behavior diagrams, Class diagrams that belong to the structure diagrams and sequence diagrams, which could be categorized as an interaction diagram (a subset of behavior diagrams).

For the creation of our UML diagrams, we utilised draw.io, a widely adopted and user-friendly online diagramming tool known for its support for a broad range of diagram types.



draw io: draw.io is a cross-platform graph drawing software application developed in HTML5 and JavaScript. Its interface can be used to create diagrams such as flowcharts, wireframes, UML diagrams, organizational charts, and network diagrams [6]. We chose draw.io for our project for its ease of use, seamless group collaboration and its wide range of diagramming features, allowing us to clearly model the system architecture and user flows.

2.3.2 Identification of the actors

The system is designed around two primary types of users:

- **Patient:** This refers to any individual who uses the system to monitor their health. Patients can be further divided into several categories:
 - Type 1 patients
 - Type 2 patients
 - Patients with gestational diabetes
 - Prediabetic patients
- **Doctor:** This actor includes different doctor's specialties who provide medical services to these patients.

2.4 Use case diagrams

A use case diagram is a graphical depiction of a user's possible interactions with a system [7]. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well. The use cases are represented by either circles or ellipses. The actors are often shown as stick figures.

2.4.1 Patient use case diagram

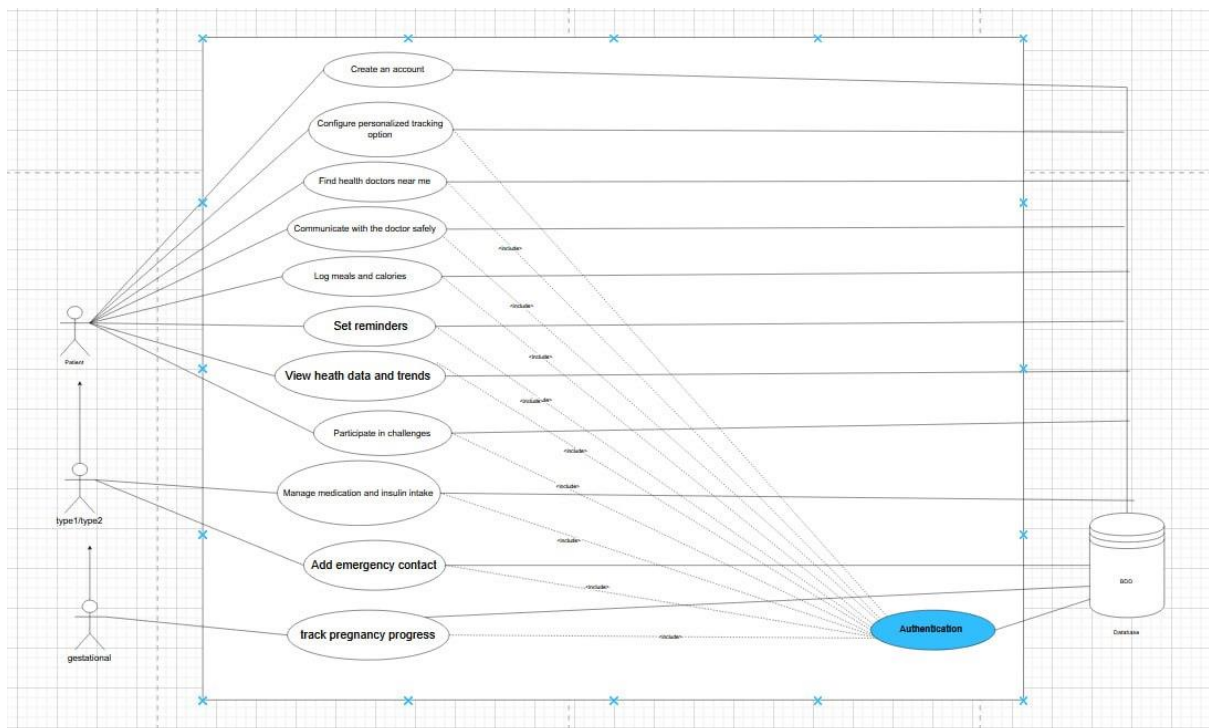


Figure 4 : Patient use case diagram.

2.4.2 Textual description of patient's use case diagram

❖ Use Case: Create an account

The use case	Create an account
Objective	Enable users to create personalised accounts and access the application's features.
Actor(s)	Patient, doctor
Pre-conditions	The user does not have an existing account beforehand
Success post-conditions	The user is successfully registered and can log in to their account
Failure post-conditions	Failure to create an account (invalid data, email already used.)
Nominal sequencing	1/ The user initiates the account creation process 2/ The system displays a registration form 3/ The user fills the form with their personal information 4/ The system validates the provided information 5/ The system registers the user's data and creates the account
Alternative sequencing	2.1/ User cancels the account creation before final confirmation 5.1/ If the information provided is incorrect, an error message would pop up requesting corrections

Table 1: Use case: create an account

❖ Use case: Monitor blood sugar

The use case	Monitor blood sugar
objective	Enable all patients to log in and track their blood sugar levels throughout the day for health monitoring purposes
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account and has the manual input option
Success post-conditions	The patient's blood sugar levels are successfully recorded and stored in the system
Failure post-conditions	Blood sugar levels are not recorded due to data input errors or system connectivity issues.
Nominal sequencing	1/ The user navigates to the blood sugar monitoring section 2/ The user manually enters their blood sugar level 3/ The system validates and stores the data 4/ The user now is able to view past records and trends
Alternative sequencing	3.1/ If the entered blood sugar level is outside a safe range, the system alerts the user and suggests contacting his doctor

Table 2: Use case: Monitor blood sugar

❖ Use case: Configure personalized tracking options

Use case	Configure personalised tracking options
objective	Enable users to customise which health parameters (e.g., weight, hydration, sleep, emotional status) they wish to track based on their individual needs
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active and logged-in account.

Success post-conditions	The system saves the user's chosen tracking preferences
Failure post-conditions	The tracking options are not saved due to a system error or database connectivity issue
Nominal sequencing	1/ The user navigates to the tracking options section 2/ The user selects the health metrics they want to track 3/ The system saves the preferences 4/ The user can now modify the registered setting anytime they want
Alternative sequencing	3.1/ If the system fails to save the selected parameters, an error message is displayed, prompting the user to try again.

Table 3: Use case: Configure personalized tracking options

❖ **Use case:** Find doctors near me

Use case	Find doctors near me
Objective	Enable patients to locate and connect with nearby doctors.
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account, and location services are enabled on their device
Success post-conditions	The patient finds and contacts a doctor near them
Failure post-conditions	No doctors matching the search criteria are found, or the system cannot access the network due to connectivity problems
Nominal sequencing	1/ The user navigates to the search section 2/ The user enters search criteria 3/ The system returns a list of matching results 4/ The user selects a doctor and can view contact details or schedule an appointment with the selected doctor.
Alternative sequencing	3.1/ If no doctors are found, the system suggests alternative search terms

Table 4: Use case: Find doctors near me

❖ **Use case:** Communicate with the doctor safely

Use case	Communicate with the doctor safely
Objective	Enable patients to securely message and communicate with their assigned doctors for consultations and health-related inquiries
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account and is already assigned to a doctor
Success post-conditions	The doctor successfully receives the message and can respond and answer to the patient's concerns
Failure post-conditions	The message fails to send due to network issues
Nominal sequencing	1/ The user navigates to the doctor communication section 2/ The user selects their assigned doctor 3/ The user types and sends a message 4 The system delivers the message to the doctor 5/ The doctor receives the sent message and responds to it

Alternative sequencing	2.1/ If no doctor is assigned to the patient, the system prompts the patient to select a doctor from a list 4.1/ If the message fails to send, an error message pops up
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Table 5: Use case: Communicate with the doctor safely

❖ **Use case:** Participate in challenges

Use case	Participate in challenges
Objective	Encourage users to stay motivated by joining health-based challenges with other users
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account and is willing to participate and meets the challenge criteria
Success post-conditions	The user is now successfully accepted in a challenge and can track their progress as well as others
Failure post-conditions	The challenge enrolment fails due to system errors (e.g., server downtime, database connection issues, or API failures)
Nominal sequencing	1/ The user navigates to the challenges section 2/ The user browses and selects a challenge that they see fits 3/ The system verifies the requirements/eligibility 4/ The user is now a participant in the challenge, and so the tracking begins 5/ The user completes challenge milestones and keeps track of progress
Alternative sequencing	3.1/ If the user is ineligible, the system suggests alternative challenges.

Table 6: Use case: Participate in challenges

❖ **Use case:** Enter a meal and track calories

Use case	Enter a meal and track calories
Objective	Allow users to log meals and calculate calorie/carb intake for effective diet management
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account
Success post-conditions	The meal is successfully logged, and the system calculates nutrient intake.
Failure post-conditions	The meal is not saved due to missing information or a system error
Nominal sequencing	1/ The user navigates to the meal tracking section 2/ The user enters or selects food items from a pre-selected database 3/ The system calculates total calories and carbs 4/ The user saves the meal log and can now review daily intake
Alternative sequencing	2.1/ If the food item is not found in the database, the user can manually enter their own

Table 7: Use case: Enter a meal and track calories

❖ **Use case:** Set reminders

Use case	Set reminders
Objective	Allow all users to schedule reminders for medications, insulin intake, meals, exercise, and blood sugar checks
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account
Success post-conditions	The reminder is successfully saved and triggers a notification at the scheduled time
Failure post-conditions	The reminder is not saved due to a system error
Nominal sequencing	1/ The user navigates to parameters, the reminders section 2 The user selects the type of reminder they want to trigger 3/ The user then selects the time and frequency 4/ The system saves and schedules the reminder type 5/ The user would receive notifications when the reminder is triggered
Alternative sequencing	3.1/ If no action is taken after the reminder, the system sends a follow-up alert

Table 8: Use case: Set reminders

❖ **Use case:** Visualise data

Use case	Visualise data.
Objective	Provide users with charts and graphs to track their blood sugar levels, meal intake, exercise and medication intake
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account and has logged at least some health data to be visualised
Success post-conditions	The user views data visualisations with insights
Failure post-conditions	The system fails to generate visuals due to missing or invalid data
Nominal sequencing	1/ The user navigates to the data visualisation section 2/ The system processes recorded data 3/ The user selects the type of data they want to visualise 4/ The system generates graphs/charts 5/ The user can analyse insights and trends
Alternative sequencing	3.1/ If no data exists, the system prompts the user to log information

Table 9: Use case: Visualise data

❖ **Use case:** Track medication/insulin intake

Use case	Track medication/insulin intake
Objective	Allow patients to log and track medication/insulin usage
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account
Success post-conditions	Medication intake is recorded
Failure post-conditions	The entry fails to save
Nominal sequencing	1/ The user navigates to the medication tracking section 2/ The user logs the dosage of medication and the time when taken 3/ The system saves the entry and updates the data

Alternative sequencing	3.1/ If the entry is invalid, an error message pops up
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Table 10: Use case: Track medication/insulin intake

❖ **Use case:** Add emergency contact

Use case	Add emergency contact
Objective	Allow users to add a trusted contact for emergency notifications
Actor(s)	Patient (type1/ type2/gestational)
Pre-conditions	The user has an active account
Success post-conditions	The contact is saved and can be notified in case of a medical emergency
Failure post-conditions	The emergency contact is not saved due to incomplete/invalid details
Nominal sequencing	1/ The user navigates to the emergency contact section 2/The user enters the contact details (name, phone, relationship) 3/ The system verifies and saves the contact in the application
Alternative sequencing	3.1/ If the contact details are incomplete or invalid, the system prompts corrections

Table 11: Use case: Add emergency contact

❖ **Use case:** Track pregnancy progress

Use case	Track pregnancy progress
Objective	Allow gestational diabetes patients to monitor their pregnancy's progression and health
Actor(s)	Gestational patients
Pre-conditions	The user has an active account
Success post-conditions	The pregnancy's data (weight, glucose levels...) is recorded and updated
Failure post-conditions	The pregnancy data fails to save due to an error
Nominal sequencing	1/the user navigates to the pregnancy tracking section 2/the user logs weight, glucose levels and probable symptoms 3/the system stores data 4/the user can now track the pregnancy's progress
Alternative sequencing	3.1/if data entry fails, the system would notify the user

Table 12: Use case: Track pregnancy progress

2.4.3 Doctor use case diagram



Figure 5 : Doctor use case diagram.

2.4.4 Textual description of doctor use cases

❖ Use Case: Create a professional profile

The use case	Create a professional profile
Objective	Enable healthcare providers to create a verified professional profile that patient interactions
Actor(s)	Doctor
Pre-conditions	The user must be a certified medical professional and provide valid documents (license or certificate)
Success post-conditions	The profile is registered and visible for patients
Failure post-conditions	Profile creation fails due to invalid data, missing details. . .
Nominal sequencing	1/ The user requests to create a professional profile 2/ A registration form is displayed 3/ The user provides personal details and uploads verification documents 4/ The system verifies the information and documents provided by the user 5/ The system registers the data. The profile is activated and published
Alternative sequencing	4.1/ If the information provided is incorrect, an error message would pop up requesting the user to upload the correct documents

Table 13: Use Case: Create a professional profile

❖ Use Case: Receive/Send alerts and notifications

The use case	Receive/Send alerts and notifications
Objective	Facilitate real-time communication between providers and patients via alerts (appointments, emergencies, ...)
Actor(s)	Doctor
Pre-conditions	The provider must have an active account

Success post-conditions	The user successfully receives and/or sends alerts
Failure post-conditions	Alerts fail due to system errors or network issues
Nominal sequencing	1/ The system generates a notification (new appointment, emergency request) 2/ The provider receives an alert 3/ The provider responds or schedules an appointment
Alternative sequencing	3.1/ If the provider is unavailable, the provider may set an automated response

Table 14: Use Case: Receive/Send alerts and notifications

❖ **Use Case: Manage patients' files**

The use case	Manage patients' files
Objective	Allow doctors to securely access, update and store patient medical records
Actor(s)	Doctor
Pre-conditions	The doctor must have an account and be authorised to view/edit the patient's file
Success post-conditions	The patient's file is updated successfully and remains secure and safe
Failure post-conditions	Data is not saved in the patient's file due to technical errors
Nominal sequencing	1/ The doctor logs into the system 2/ The doctor searches for a specific patient record 3/ The system retrieves and shows the file 4/ The doctor reviews and edits necessary details (diagnoses, prescriptions, progress notes) 5/ The system saves the changes
Alternative sequencing	2.1 if access is restricted, the doctor requests administrator approval

Table 15: Use Case: Manage patients' files

❖ **Use Case: Manage appointments**

The use case	Manage appointments
Objective	Enable doctors to schedule, modify, or cancel appointments
Actor(s)	Doctor
Pre-conditions	The doctor must have an active account
Success post-conditions	Appointments are updated and confirmed
Failure post-conditions	The appointment request fails due to conflicts or system errors
Nominal sequencing	1/ The doctor checks the appointment dashboard 2/ They check availability and schedule an appointment accordingly 3/ The system confirms and notifies both parties 4/ If a change is needed, the parties involved reschedule or cancel the appointment 5/ The system updates the appointment dashboard records accordingly
Alternative sequencing	4.1/ The patient declines the appointment; they are prompted to choose a new slot

Table 16: Use Case: Manage appointments

❖ **Use Case: Post educational articles**

The use case	Post educational articles
Objective	Allow doctors to share knowledge and research findings through articles
Actor(s)	Doctor
Pre-conditions	The doctor must have an active professional profile and article posting privileges
Success post-conditions	The article is published and accessible to users
Failure post-conditions	The article is not posted due to system errors or moderation flags
Nominal sequencing	1/ The doctor accesses the post article section 2/ They compose or upload an article 3/ The system checks for guidelines compliance 4/ Once approved, the article is published
Alternative sequencing	3.1/ If the article gets flagged, the provider receives feedback for revision

Table 17: Use Case: Post educational articles

❖ **Use Case:** Give medical guidance and prescriptions

The use case	Give medical guidance and prescriptions
Objective	Enable doctors to provide advice and issue prescriptions based on the patient's records
Actor(s)	Doctor
Pre-conditions	The doctor should have an active account and authorised access to the patient's medical history
Success post-conditions	The patient receives guidance or a prescription
Failure post-conditions	The prescription fails to generate due to technical issues or missing data
Nominal sequencing	1/ The doctor accesses the patient's medical record 2/ The doctor provides medical advice or prescribes medication after studying the symptoms 3/ The system securely saves and shares the advice/prescription with the patient
Alternative sequencing	3.1/ If further tests are needed, the doctor requests them before prescribing any type of medication

Table 18: Use Case: Give medical guidance and prescriptions

❖ **Use Case:** create challenges

The use case	Create challenges
Objective	Enable doctors to design health challenges for patient engagement
Actor(s)	Doctor
Pre-conditions	The doctor must have an active profile
Success post-conditions	The challenge is published and available for participation
Failure post-conditions	The challenge is not published due to missing information
Nominal sequencing	1/ The doctor navigates to the challenge section 2/ The doctor defines challenge goals and rules 3/ The system publishes the challenge and notifies participants

Alternative sequencing	3.1/ If guidelines are unmet the doctor would be asked to modify it and submit a revised version
------------------------	--

Table 19: Use Case: create challenges

❖ **Use Case:** Answer patient FAQ

The use case	Answer patient FAQ
Objective	Allow doctors to respond to common patient inquiries and provide professional medical guidance
Actor(s)	Doctor
Pre-conditions	The doctor must have an active account, and the FAQ section must be accessible
Success post-conditions	The patients' questions are answered and published in the FAQ section
Failure post-conditions	The response is not submitted due to incomplete data or a system error
Nominal sequencing	1/ A doctor submits a question in the FAQ section 2/ The doctor accesses the question via the FAQ section and reviews its details 3/ The doctor writes and submits an answer 4/ The system verifies that the response is clear and meets the guidelines 5/ The response is published and visible to patients
Alternative sequencing	4.1/ If the response contains unclear or incomplete information, then the provider is prompted to revise it

Table 20: Use Case: Answer patient FAQ

2.5 Class diagram

The class diagram below (Figure 4) illustrates the relationships between the main components of the diabetes management system, including patients, doctors, health data, and more, providing a clear overview of the application's architecture.

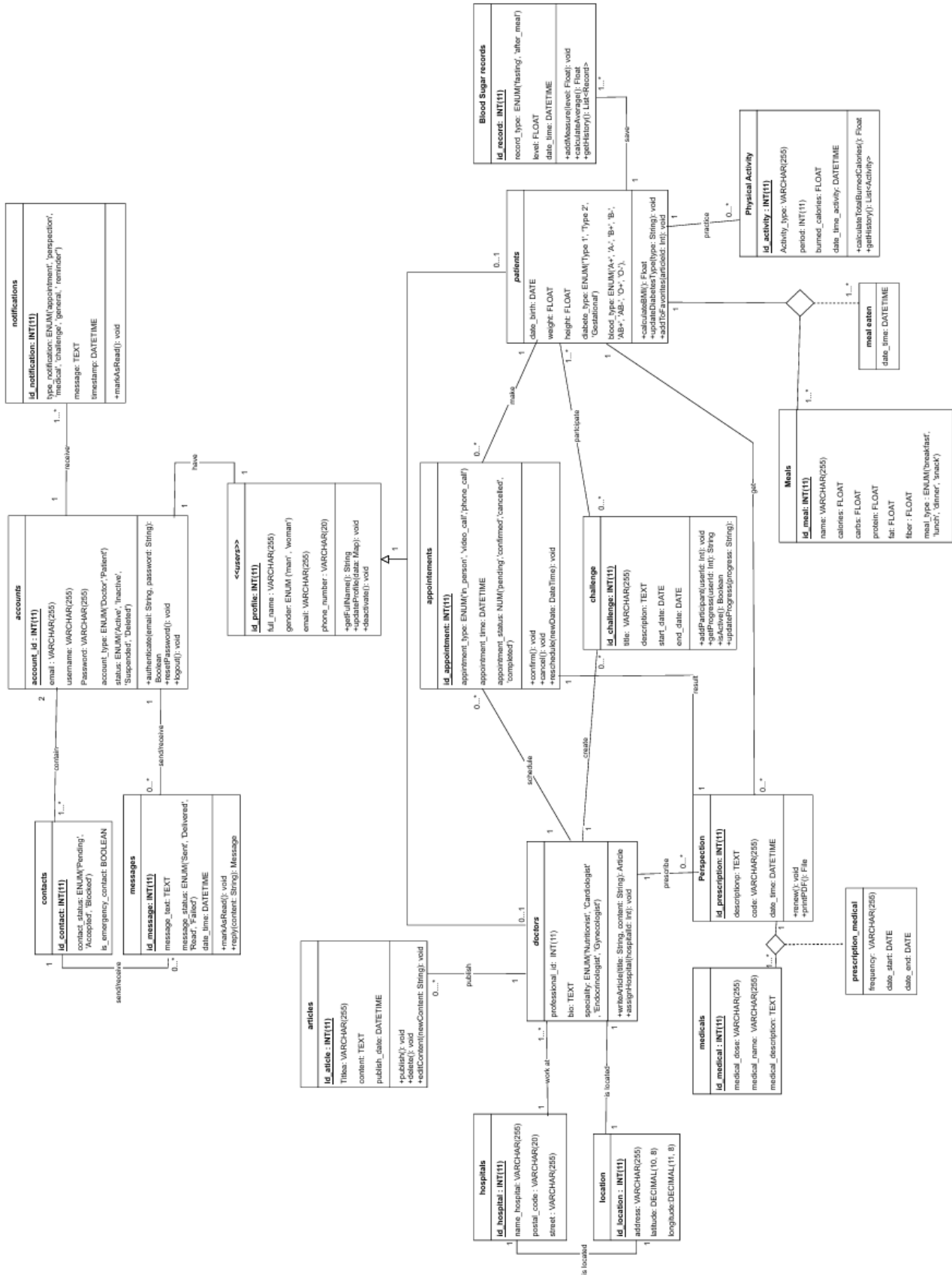


Figure 6 : Class diagram

2.6 Sequence diagrams

The sequence diagrams below (figures 4 to 7) focus primarily on user account interactions, such as registration and login. In addition, we have included a diagram for the glucose tracking process, which represents one of the system's key functionalities.

2.6.1 Account creation sequence diagram

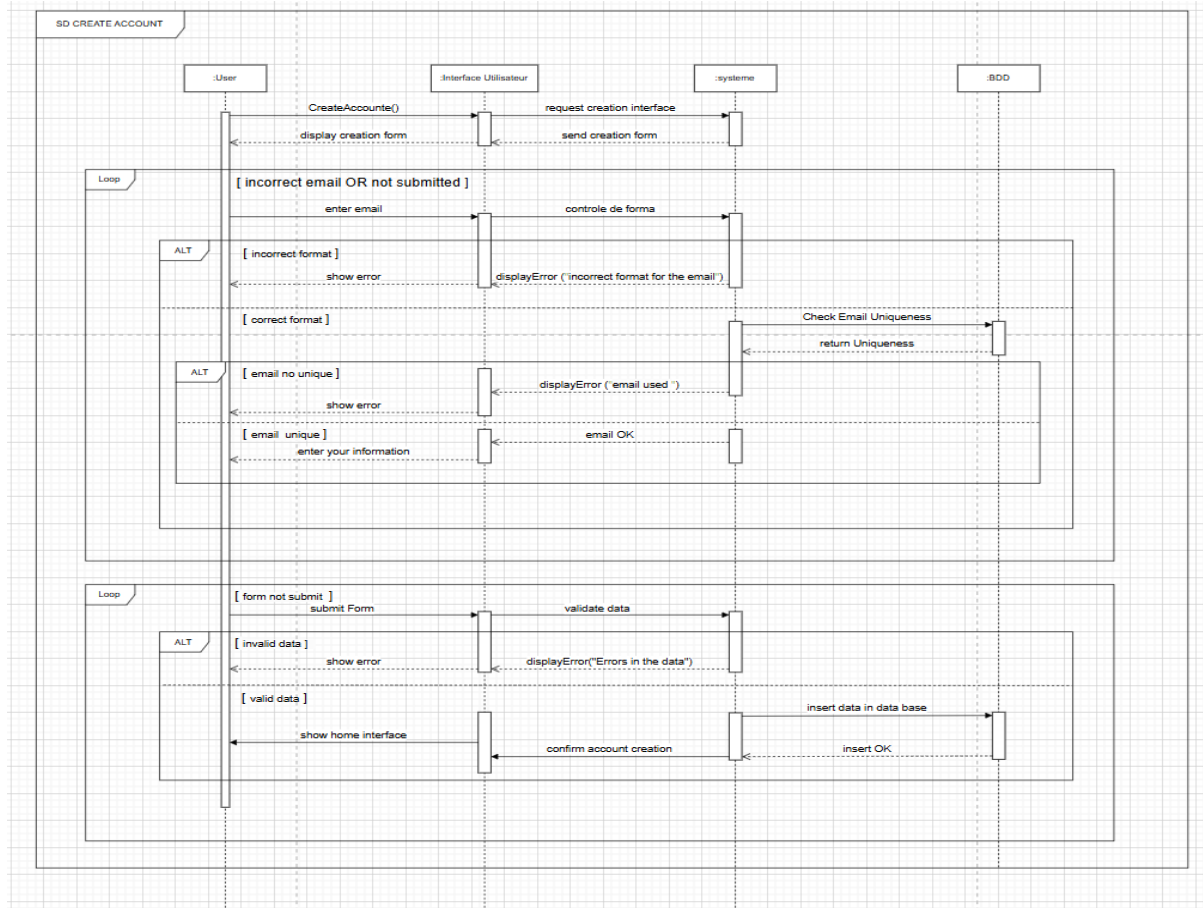


Figure 7 : Account creation sequence diagram

2.6.2 Authentication sequence diagram

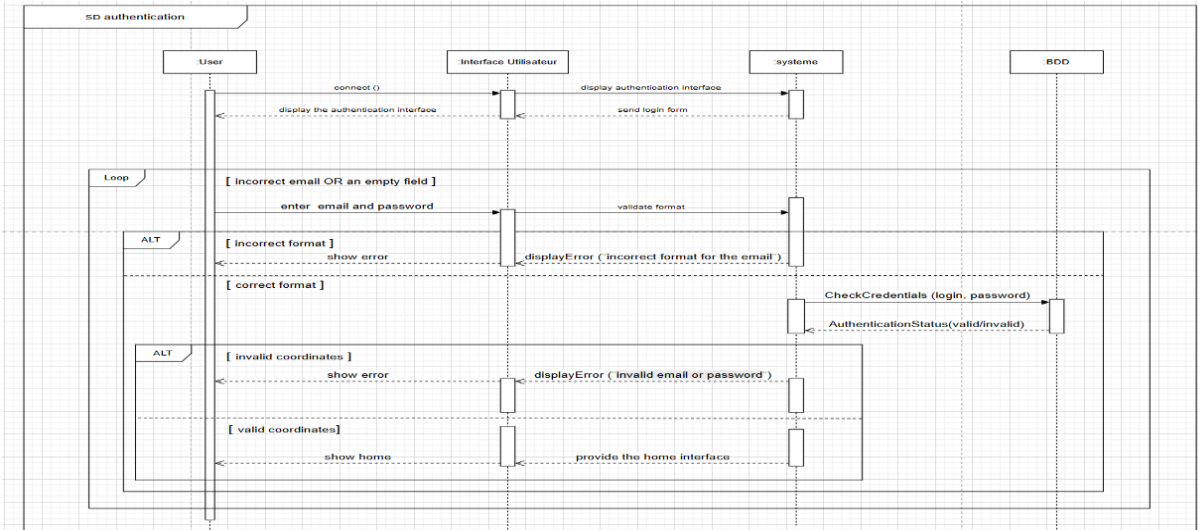


Figure 8 : Authentication sequence diagram.

2.6.3 Glucose logging and monitoring sequence diagram

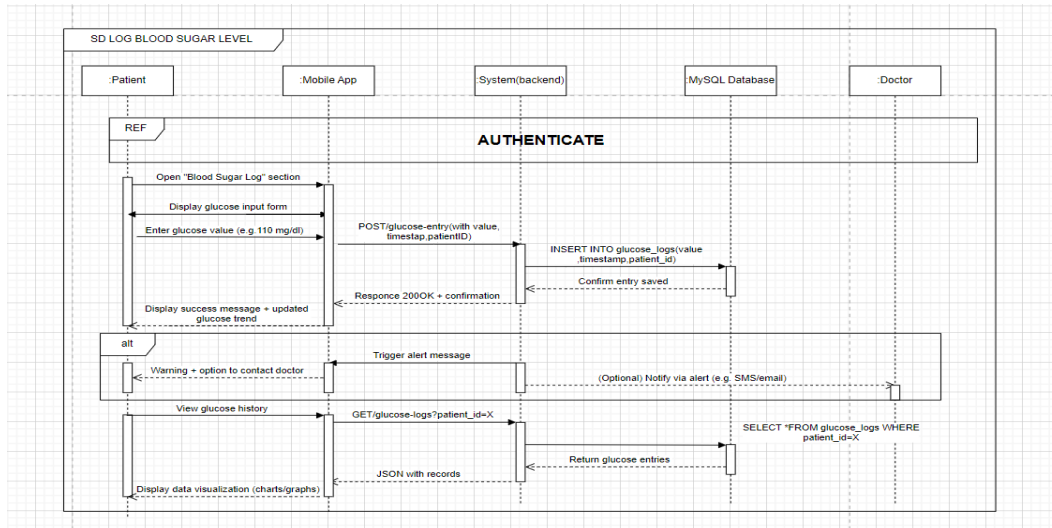


Figure 9 : Glucose logging and monitoring sequence diagram.

2.7 Conclusion

This chapter outlined all the essential requirements and design components of our diabetes management system. We defined both functional and non-functional specifications, identified key actors, and illustrated user interactions via use case, class and sequence diagrams. These models would provide a firm foundation for understanding not only the system's structure but also its overall behavior. With this basic groundwork, the next chapter will focus on the actual implementation of the proposed solution, detailing the modules, technologies, and development process involved.

Chapter 3: Implementation of the Proposed Solution

3.1. Introduction

This chapter presents the global software and technological environment adopted for the development of the diabetes monitoring application. It describes the processes of selecting the operating systems, programming languages, frameworks, and tools used during the implementation phase. This section also explains the use of APIs for communication between system components and outlines the security measures applied to protect sensitive medical data. Finally, various user interfaces developed after coding are showcased to illustrate the application's structure and functionality.

3.2. Development Tools and Environments

The development of our mobile application relies on a combination of modern tools and environments. **GitHub** was used for team coding collaboration, **Visual Studio Code** served as the primary code editor, and **MySQL** was chosen to manage structured data.

3.2.1 GitHub



GitHub is a cloud-based platform that provides version control, code hosting, and collaboration tools for software development. It is built around Git, a distributed version control system created by Linus Torvalds. It is the most widely used platform for both individual and team-based development, supporting open-source and private projects alike. [9]

For our application, GitHub played a vital role in organizing, securing, and managing the entire codebase, including frontend and backend codes as well as database scripts, while enabling effective team collaboration.

3.2.2 Visual Studio Code (VS Code)



Visual Studio Code is a source-code editor developed by Microsoft. It is lightweight, fast, and highly extensible through a wide range of plugins and extensions. It supports many programming languages including Dart (for Flutter) and Python (for backend development). [10]. In our application, VS Code serves as the primary development environment for writing both the backend and frontend codes in Python and Dart respectively.

3.2.3 MySQL



MySQL is an open-source Relational Database Management System (RDBMS) based on Structured Query Language (SQL). It is widely used to store and manage large amounts of data in tables made up of rows and columns for various applications and websites. This system is known for its speed, reliability, and ease of use. It also supports indexing, querying, and efficient data retrieval, making it a strong choice for applications that handle structured and interconnected data. [11]. In this project, MySQL is used as the central database that securely stores and organizes all user's medical data and related information. It plays a crucial role in managing the persistence layer of the system and ensures the consistency and integrity of health records.

3.3. Operating Systems: Android & iOS

Our application is designed as a cross-platform solution, supporting both Android and iOS operating systems. This ensures that users with different mobile devices can benefit from the application's features seamlessly.

3.3.1 Android Operating System



Android is an open-source mobile operating system developed by Google, based on the Linux kernel. It powers a wide range of smartphones and tablets from various manufacturers including Samsung, Xiaomi, and Huawei [12]. The application is intended to run on Android devices, making Android the target platform for development and testing.

3.3.2 iOS Operating System



iOS is a proprietary mobile operating system developed by Apple Inc. It runs on iPhones and iPads. It is known for its strong emphasis on security, performance, and user experience. [13] supporting iOS ensures that the diabetes management application reaches a wider audience, including Apple device users.

3.4. Frameworks

We used frameworks for building our mobile application, so we choose both flutter and flask. While Flutter is for building the frontend, offering native performance on both Android and iOS. Flask was adopted for the backend to handle the server-side logic and API services.

3.4.1 Flutter



Flutter is an open-source framework developed and supported by Google for building cross-platform applications from a single codebase. It allows developers to create high-performance, visually attractive apps for Android, iOS, web, and desktop using the Dart programming language. It uses a reactive programming model and includes its own rendering engine called Skia, enabling developers to build fully custom, native-looking user interfaces without relying on the native platform's UI components. Also has strong support for animations, custom components and responsive design, in addition to its integration with REST APIs, and local databases. [15]. In this application, we used it to develop the mobile user interface. The goal is to provide diabetic patients and doctors with a fast, responsive, and user-friendly experience on both Android and iOS devices without duplicating development efforts, it reduces code duplication and time.

3.4.2 Flask (for Python)



Flask is a lightweight and micro web framework written in Python. It is designed to be simple yet extensible, allowing developers to build secure and scalable web applications quickly. Flask follows the Web Server Gateway Interface (WSGI) standard. [16] [17]. In this project, Flask is used to create the RESTful APIs that serve as the communication bridge between the mobile frontend (Flutter) and the backend services which interact with the MySQL database. It handles user authentication, receives and saves glucose readings or meals entries, and manages secure data flow between components.

3.5. Programming languages

The development involved two main programming languages: Dart, used within Flutter for the frontend, and Python, utilized in building the backend services and handling data processing tasks.

3.5.1 Dart



Dart is a client-optimized, object-oriented programming language developed by Google. [8] It is designed to build fast and scalable web and mobile applications. Dart is the primary language used in Flutter.[14] [18]. It is used to develop the mobile user interface and logic (frontend). It handles user interactions, data input (such as blood sugar readings or meals), and communicates with the backend via HTTP APIs. Its syntax is simple and clean, making it easy to maintain and debug.

3.5.2 Python Programming Language



Python is a high-level, interpreted programming language known for its readability, versatility, and strong ecosystem. It is widely used in web development, data science, machine learning, and artificial intelligence. [19]. In this project, Python was used to build the backend of the application. It powers the API that connects the mobile app to the MySQL database. Python's simplicity and extensive libraries make it ideal for rapid backend development.

3.6. Application Programming Interface (API) and Integration



In this project, an API serves as the communication bridge between the mobile application developed with Flutter and the backend server built with Python (Flask). It follows the REST (Representational State Transfer) architecture, allowing the application to securely send and receive data such as user information, glucose levels, and meal records. By using HTTP requests (GET, POST, PUT, DELETE), the API supports seamless communication between the frontend and backend while ensuring data consistency.

3.7. Security Measures



To protect sensitive medical information and ensure user privacy, the system implements essential security measures. These are authentication mechanisms to verify user identity, role-based access control to restrict functionalities according to user type (patients or doctors), and secure data transmission through HTTPS protocol. These features help safeguard health information against unauthorized access and preserve data confidentiality throughout the application.

3.8. Application's Interfaces

This section showcases the finalized user interface screens, highlighting the main features available to both patients and doctors. It includes visual representations of key functionalities such as health monitoring, meal tracking, and personalized dashboards. These interfaces were designed to provide a seamless and intuitive user experience across both Android and iOS platforms.

3.8.1 Application's name and logo

We named our application **Sukari**, inspired by the Arabic word for “diabetes to highlight our focus on blood sugar management and supporting individuals managing diabetes, the logo features a blood testing device, symbolizing the absolute essential role of blood sugar monitoring in effective diabetes care.



Figure 10 :App Logo.

3.8.2 Representation of interfaces

Below are examples of the primary screens implemented in the application, demonstrating core functionalities and visual layout, adapted for different user roles.

3.8.2.1 Login page and Sign up pages for both users

The Sign In page features the Sukari logo at the top, followed by the tagline "Sweet support for a balanced life". Below this is a "Sign In" section with input fields for "Email" and "Password". A green "LOGIN" button is positioned below the password field. At the bottom, there are two buttons: "Register as Patient" and "Register as Doctor".

Figure 11 : Sign in page.

The Patient sign up page displays the Sukari logo and tagline. The main heading is "REGISTER AS PATIENT" with the subtext "Create your account to get started". The form includes input fields for "Full Name", "Email", "Phone Number", and "Date of Birth" (with a calendar icon). It also features a "Gender" dropdown menu (currently set to "Male") and a "Country" dropdown menu.

Figure 12 : Patient sign up.

The Doctor sign up page shows the Sukari logo and tagline. The main heading is "REGISTER AS DOCTOR" with the subtext "Create your doctor account to get started". The form includes input fields for "Full Name", "Email", "Phone Number", and "Medical Specialty" (with a dropdown menu). It also features a "Clinic/Hospital Name" input field and a "Country" dropdown menu.

Figure 13 : Doctor sign up.

3.8.2.2 Patient's interface

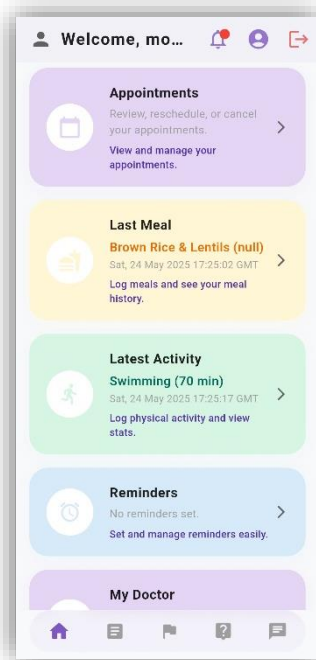


Figure 14 : Patient home page.

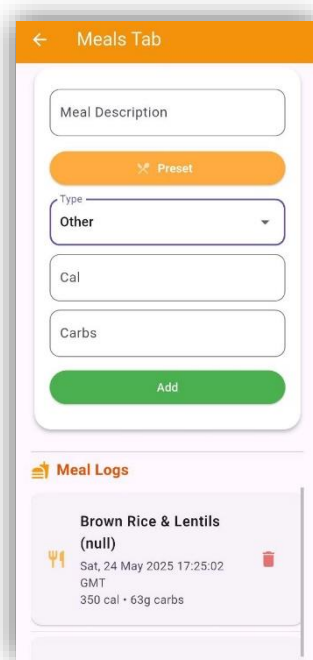


Figure 15 : Meal tracking tab.

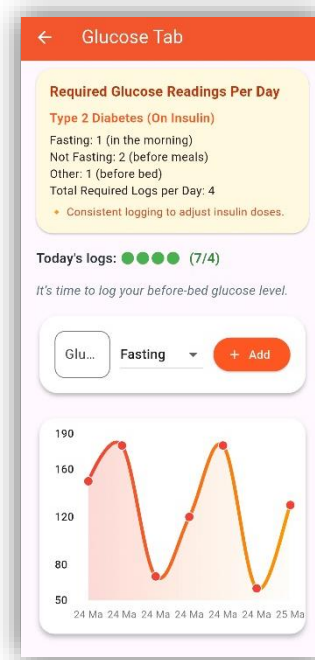


Figure 16 : Glucose tracking tab.

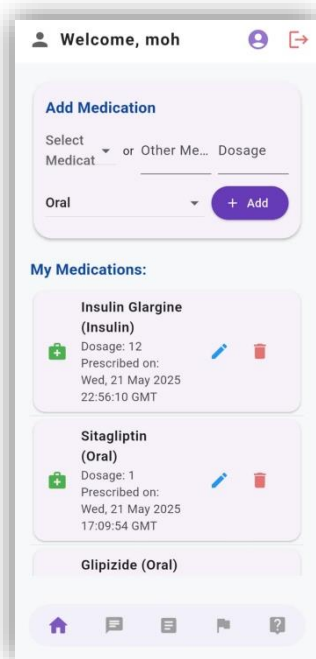


Figure 17 : Medication tracking tab.

3.8.2.3 Doctor's interface

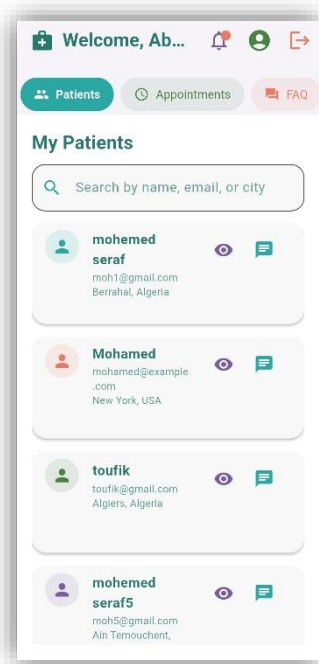


Figure 18: Doctor home page.

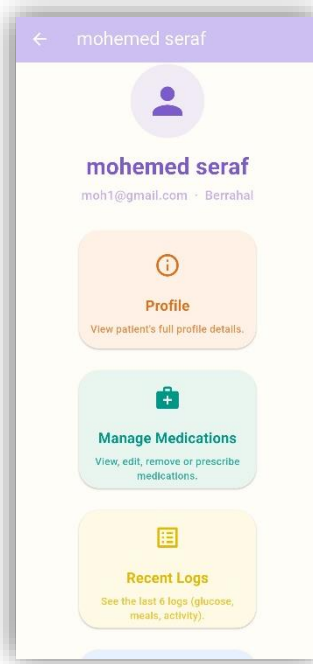


Figure 19 :Patient management page.

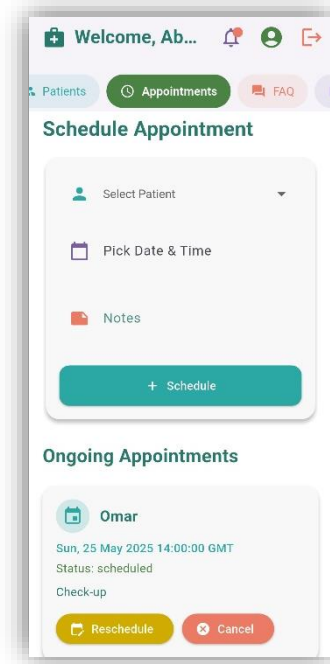


Figure 20 : Appointment tab.

3.9 Conclusion

In this chapter, we described the technological choices and implementation process for our diabetes monitoring application. We detailed the selection and integration of development tools, programming languages, frameworks, and security measures, as well as the design of user interfaces. Throughout the implementation phase, we encountered several challenges, particularly in ensuring seamless communication between the frontend and backend, and in maintaining data security.

Despite these challenges, the core functionalities of the application were successfully developed, resulting in a stable and user-friendly system that meets the initial project requirements, tailored for both diabetic patients and doctors.

General Conclusion

Working on this thesis has been an incredibly rewarding and enriching experience for our team, providing valuable lessons that will benefit us both personally and professionally. This project gave us the opportunity to apply our technical skills to a meaningful real-world problem while deepening our understanding of the challenges faced by people living with diabetes and helping develop a genuine connection with them. Although we encountered a few difficult moments, bugs, and unexpected mishaps along the way, each obstacle contributed to our growth, improved our collaboration, and strengthened our problem-solving and teamwork skills.

Our application offers essential features such as blood sugar monitoring, follow-up reminders with doctors, and specialised interfaces tailored to different medical professionals. By designing the app to meet the specific needs of users, whether they are managing type 1, type 2, prediabetes, or gestational diabetes, we aimed to create a practical and supportive health tool.

Looking ahead, we plan to introduce new features such as a community platform to help patients better adapt to their condition, synchronization with wearable devices. In addition, we aim to integrate AI-based nutritional analysis to provide personalized dietary recommendations. These enhancements are intended to better address the specific needs of our users and make diabetes management as effective and accessible as possible. Furthermore, we are committed to continuously improving the application based on user feedback after launch to ensure the best possible user experience and facilitate more effective health monitoring.

This thesis marks an important step in our journey. We are excited about the future possibilities for expanding and enhancing our project and are deeply grateful for the opportunity to work on such a meaningful and impactful application. This work represents just the beginning of what we hope will evolve into a powerful tool for diabetic care and prevention.

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